BDI-Grounded Argumentation for Multi-Perspective AI Policy Analysis: Integrating QBAF, FIRE, and Ontological Framing in a Living Taxonomy

Abstract

AI policy discourse involves competing empirical claims, normative commitments, and strategic reasoning that existing NLP tools compress into lossy single-label representations. We present an integrated system for multi-perspective discourse analysis that combines three layers: (1) ontological grounding using a composite of DOLCE D&S (perspectival multiplicity), BDI (Belief-Desire-Intention agent characterization), and AIF (Argument Interchange Format) vocabulary; (2) formal argumentation via Quantitative Bipolar Argumentation Frameworks (QBAFs) with a novel BDI-aware base score calibration that addresses the fundamental asymmetry between empirical and normative claim assessment; and (3) confidence-gated iterative claim extraction (FIRE) that replaces single-shot summarization with a per-claim verification loop guided by evidence criteria heuristics. Our system organizes AI policy literature through a four-POV taxonomy (accelerationist, safetyist, skeptic, and shared situations) with ~500 nodes, supports multi-agent debates with ontology-grounded context injection, and feeds debate findings back into the taxonomy through concession harvesting. We evaluate QBAF base score calibration across BDI layers, finding that AI reliably scores normative (Desires, r=0.65) and strategic (Intentions, r=0.71) claims but not empirical (Beliefs) claims — a fundamental asymmetry we attribute to the self-contained vs. externally-verifiable nature of different claim types. We describe FIRE's extraction methodology and present preliminary results on BDI-structured vs. flat context injection for debate quality. [RESULTS PENDING for E1, E3]

1. Introduction

The rapid development of AI systems has produced a complex policy discourse involving multiple stakeholder communities with fundamentally different analytical frameworks. Accelerationists emphasize scaling and capability growth; safetyists prioritize alignment and existential risk mitigation; skeptics focus on immediate, measurable harms like bias and labor displacement. These communities frequently disagree not because they have different information, but because they operate from different normative commitments, define key terms differently, and employ different reasoning strategies.

Current NLP tools for discourse analysis — stance detection, topic modeling, sentiment analysis — struggle with this multi-dimensional complexity. They project what are fundamentally different kinds of disagreement (empirical vs. values-based vs. definitional) onto flat, single-label representations that lose the structure needed to make policy disagreements tractable.

1.1 The Projection Problem in Stance Detection

Consider a claim like "AI regulation should be based on demonstrated harm, not hypothetical risk." Standard stance detection classifies this as either *Favor* or *Against* AI regulation. But this classification compresses at least three independent dimensions: (1) a factual belief about how regulation works best (empirical), (2) a normative commitment to proportionality over precaution

(values), and (3) a strategic preference for reactive over proactive governance (methodology). Two annotators who agree on all three dimensions but weight them differently will produce different labels, generating spurious inter-annotator disagreement.

This projection problem (Ebrahimi et al., 2022) is not a labeling error — it is a structural limitation of binary stance representations applied to multi-dimensional policy opinions. When LLMs are used for stance classification, they inherit the same compression artifact, producing systematically unreliable labels on exactly the claims where nuance matters most.

1.2 Contributions

We present a system that addresses the projection problem through three integrated layers:

1. **Ontological grounding.** A composite ontology using DOLCE D&S for perspectival multiplicity (multiple descriptions of shared situations), BDI for agent characterization (separating empirical beliefs from normative desires and strategic intentions), and AIF for argumentation structure (typed attack/support relationships with formal scheme vocabulary). We adopt ontological *vocabulary* in JSON structures rather than OWL/RDF formal reasoning — a design choice we term "vocabulary over formalism."

2. **Formal argumentation with BDI-aware scoring.** A QBAF implementation using DF-QuAD gradual semantics (Rago et al., 2016) with a novel BDI-aware base score calibration. Through iterative calibration (Q-0), we discovered that different BDI layers require fundamentally different scoring approaches: normative and strategic claims can be AI-scored via decomposed evidence criteria, while empirical claims require human judgment — a finding we attribute to the self-contained vs. externally-verifiable nature of different claim types.

3. **Confidence-gated iterative extraction (FIRE).** A replacement for single-shot claim extraction that assesses per-claim confidence through evidence criteria heuristics and iteratively refines uncertain claims through targeted follow-up queries. FIRE addresses three identified failure modes of single-shot extraction: specificity collapse, warrant deficit, and claim clustering.

The system is deployed and actively used for AI policy research, operating on ~80 source documents, ~500 taxonomy nodes, ~1,500 conflict instances, and ~25 structured debate sessions.

2. Related Work

2.1 Argument Mining and Claim Extraction

Argument mining has progressed from identifying argumentative discourse units (Stab and Gurevych, 2017) to end-to-end claim extraction and relation classification (Lauscher et al., 2022; Mayer et al., 2020). Recent work leverages LLMs for claim extraction (Törnberg, 2024), achieving strong performance on standard benchmarks. However, most argument mining pipelines stop at extraction — they identify claims and classify attack/support relationships but do not compute formal argument strength or propagate that strength through an argument network.

Our system extends the extraction pipeline with two innovations: (1) FIRE adds confidence-gated verification to the extraction step, distinguishing reliably extracted claims from potential

hallucinations; and (2) QBAF integration propagates base scores through the extracted argument network via DF-QuAD gradual semantics, producing computed strength values that reflect the full attack/support context rather than isolated claim quality.

2.2 Stance Detection and Multi-Dimensional Opinion

Binary stance detection (Mohammad et al., 2016; ALDayel and Magdy, 2021) classifies text as Favor/Against a target. While effective for simple targets, binary classification fails on complex policy topics where opinions span multiple independent dimensions. Recent work acknowledges this limitation: Li et al. (2023) propose multi-dimensional stance analysis, and Luo et al. (2024) decompose stance into sub-components. However, these approaches use ad-hoc dimensional decompositions without principled grounding in argumentation theory or cognitive science.

We propose BDI decomposition (Bratman, 1987; Rao and Georgeff, 1991) as a principled alternative. BDI originates in philosophy of mind and was formalized for agent-based systems. Applied to discourse analysis, it separates empirical claims (Beliefs), normative commitments (Desires), and strategic reasoning (Intentions). This decomposition is not ad-hoc — each category has a disambiguation test: "Could this be proven true or false with evidence?" (Belief), "Is this about what ought to happen?" (Desire), "Is this about how to achieve a goal?" (Intention). The categories are exhaustive for policy discourse and mutually exclusive when disambiguation tests are applied.

2.3 Computational Argumentation Frameworks

Abstract argumentation frameworks (Dung, 1995) compute argument acceptability from attack relationships. Bipolar frameworks (Cayrol and Lagasquie-Schiex, 2005) add support relationships. Quantitative extensions assign numerical strength to arguments: QBAFs (Baroni et al., 2019) combine base scores with gradual semantics to propagate strength through attack/support networks. DF-QuAD (Rago et al., 2016) provides a specific gradual semantics that is discontinuity-free and guarantees convergence.

The Argument Interchange Format (AIF, Chesnevar et al., 2006; Rahwan et al., 2007) provides a standardized vocabulary for representing arguments: I-nodes (information — claims), S-nodes (scheme — reasoning patterns), and CA-nodes (conflict — attacks). Tools like OVA (Janier and Reed, 2014) and AIFdb (Lawrence et al., 2012) implement AIF for manual and semi-automated annotation.

Our contribution is twofold: (1) we apply QBAF to real policy discourse at scale with calibrated base scores (existing QBAF work primarily uses synthetic data or small-scale experiments), and (2) we discover and address a BDI-layer-dependent asymmetry in base score calibration that has not been reported in prior work.

2.4 Ontology-Grounded NLP

Ontologies have been used in NLP for knowledge representation (Guarino et al., 2009), entity typing (Ling and Weld, 2012), and domain-specific information extraction (Jonnalagadda et al., 2012). DOLCE (Masolo et al., 2003) has been applied to discourse analysis through its Descriptions and Situations (D&S) extension, which models how the same situation can receive different

descriptions from different perspectives. BFO (Smith et al., 2015) dominates biomedical ontology but assumes a mind-independent reality that is ill-suited to perspectival discourse analysis.

We adopt a composite ontology: DOLCE D&S provides the perspectival framework (three POV "descriptions" of shared "situations"), BDI provides agent characterization (structuring each perspective's internal reasoning), and AIF provides argumentation vocabulary (formalizing how perspectives interact). Critically, we adopt ontological *vocabulary* — naming conventions, category tests, and description patterns — rather than formal OWL/RDF triples. This "vocabulary over formalism" approach provides sufficient grounding for discourse analysis without the engineering overhead of formal ontology reasoning.

2.5 LLM-as-Debater and Multi-Agent Argumentation

Multi-agent debate using LLMs has been explored for factual accuracy improvement (Du et al., 2023), reasoning enhancement (Liang et al., 2023), and deliberative alignment (Chan et al., 2024). These systems typically assign agents fixed positions and evaluate debate outcomes on factual benchmarks.

A persistent problem in multi-agent debate is *rhetorical rigidity*: agents defend assigned stances without genuine concession, producing repetitive exchanges that fail to converge on shared understanding. Khan et al. (2024) show that allowing agents to self-select positions improves factual accuracy. Our system addresses rhetorical rigidity through three mechanisms: (1) a dialectical move taxonomy with diversity enforcement (preventing repetitive CONCEDE-DISTINGUISH cycling), (2) per-debater commitment tracking that prevents silent self-contradiction, and (3) concession harvesting that propagates genuine concessions back to the taxonomy, closing the loop between argumentation output and ontology evolution.

3. System Architecture

The system implements a five-stage pipeline: **ingest** (document conversion and metadata extraction), **extract** (claim identification with confidence assessment), **argue** (multi-agent debate with ontology-grounded context), **synthesize** (argument mapping, preference resolution, disagreement typing), and **evolve** (taxonomy updates via debate harvest, concession accumulation, and health analysis).

3.1 Ontological Grounding Layer

Our composite ontology layers three frameworks, each addressing a distinct aspect of multi-perspective discourse:

DOLCE D&S (perspectival multiplicity). Situation nodes represent contested concepts that all three POVs engage with but interpret differently. Each situation node carries three interpretation fields — one per POV — describing how that perspective understands the concept. For example, "AI Governance" is interpreted by accelerationists as a potential innovation bottleneck, by safetyists as essential gating for high-risk systems, and by skeptics as a capture risk requiring independent oversight. This structure forces debate agents to engage with how other perspectives see the same concept, preventing agents from talking past each other.

BDI (agent characterization). Each POV's taxonomy nodes are classified into three BDI categories using explicit disambiguation tests: - **Beliefs** (empirical grounding): "Could this be proven true or false with evidence?" Claims about how the world is. - **Desires** (normative commitments): "Is this about what ought to happen?" Claims about what matters. - **Intentions** (reasoning approach): "Is this about how to achieve a goal or how to reason about the issue?" Claims about strategy and methodology.

When taxonomy context is injected into debate agents, nodes are grouped by BDI category with explicit framing headers: "YOUR EMPIRICAL GROUNDING (what you take as true)," "YOUR NORMATIVE COMMITMENTS (what you argue should happen)," and "YOUR REASONING APPROACH (how you construct arguments)." This BDI-structured presentation teaches the AI its worldview structure rather than presenting a flat list of positions.

AIF (argumentation vocabulary). Seven canonical, AIF-aligned edge types formalize relationships between taxonomy nodes: SUPPORTS, CONTRADICTS, ASSUMES, WEAKENS, RESPONDS_TO, TENSION_WITH, and INTERPRETS. Attack types follow Pollock's tripartite classification: *rebut* (contradicts the conclusion), *undercut* (denies the inference), and *undermine* (attacks premise credibility). This vocabulary is used consistently across extraction, debate, and synthesis.

3.2 Taxonomy Structure

The taxonomy contains approximately 500 nodes organized into four POVs: - **Accelerationist** (prefix `acc-`): positions emphasizing AI capability growth, scaling, and transformative potential - **Safetyist** (prefix `saf-`): positions emphasizing alignment, existential risk, and deployment caution - **Skeptic** (prefix `skp-`): positions emphasizing immediate harms, bias, labor displacement, and institutional accountability - **Situations** (prefix `sit-`): shared concepts that all three POVs engage with, carrying three interpretation fields

Node descriptions follow a DOLCE-derived genus-differentia format: "A [Belief|Desire|Intention] within [POV] discourse that [differentia]. Encompasses: [what it covers]. Excludes: [boundaries]." This format enforces semantic precision — the genus (BDI category + POV) classifies the node, the differentia distinguishes it from neighbors, and the Encompasses/Excludes clauses define boundaries.

Each node carries optional graph attributes populated organically through AI-assisted extraction: `epistemic_type` (normative_prescription, empirical_claim, etc.), `rhetorical_strategy`, `node_scope` (claim/scheme/bridging, per AIF), `steelman_vulnerability` (per-POV strongest counterargument), and `possible_fallacies` (with four-tier classification: formal, informal_structural, informal_contextual, cognitive_bias).

3.3 BDI-Structured Context Injection

A critical design choice is how taxonomy context is presented to AI debate agents. Flat injection — dumping all relevant nodes as a list — causes attention dilution: the model treats all nodes equally, and instruction-following degrades as context length increases.

Our `formatTaxonomyContext` function implements three strategies to combat attention dilution:

1. **BDI grouping.** Nodes are organized into three sections with framing headers that teach the agent what each category means and how to use it. This structure prevents the agent from treating empirical facts and normative commitments interchangeably.
2. **Relevance tiering.** When embedding-based relevance scores are available (via cosine similarity against the debate topic using all-MiniLM-L6-v2 embeddings), the top-5 nodes per BDI category are marked as primary (★) with explicit instruction to prioritize them. Supporting nodes provide broader context but should not dominate the response.
3. **CHESS dynamic branch injection.** For large taxonomies, only nodes within relevant taxonomy branches are injected at full depth; other branches contribute only top-level nodes as a "safety margin." This bounds context size while preserving coverage.

4. FIRE: Confidence-Gated Iterative Claim Extraction

4.1 Motivation: Single-Shot Extraction Failures

Single-shot claim extraction — one AI call to extract all claims from a document — exhibits three systematic failure modes that degrade extraction quality:

Specificity collapse. When uncertain about a document's claims, the model defaults to vague formulations that are technically true but analytically useless: "AI has potential risks" instead of "GPT-4′s emergent capabilities in bioweapons synthesis pose proliferation risks." Vague claims are hard to attack, support, or classify — they survive extraction but provide no analytical value. Specificity collapse is especially prevalent in long documents where later sections receive less attention.

Warrant deficit. The model extracts surface-level claims (what the document says) without engaging the argument structure (why the document says it). A warranted claim connects evidence to conclusion through explicit reasoning; an unwarranted claim is a bare assertion. The warrant deficit represents a failure of comprehension — the model is keyword-matching rather than understanding.

Claim clustering. The model fixates on the most salient theme in the document and generates variations of the same claim, missing secondary arguments. This "attention tunnel" effect is especially common when the document's opening paragraphs state a strong thesis that dominates the model's attention.

4.2 The Evidence Criteria Heuristic

FIRE assesses extraction reliability through a heuristic computed from three universal criteria on each claim's `evidence_criteria`:

Criterion	Weight	What it detects
`specificity = precise`	+0.2	Claim names identifiable entities, quantities, or mechanisms
`has_warrant = true`	+0.2	Claim includes reasoning connecting evidence to conclusion
`internally_consistent = true`	+0.1	Claim doesn't contradict other extracted claims

Base confidence: 0.3. Threshold for acceptance: 0.7. Range: 0.3 (vague, unwarranted, inconsistent) to 0.8 (precise, warranted, consistent).

The heuristic captures *extraction surface quality* — observable textual properties that correlate with reliable extraction. It deliberately does not capture factual accuracy (a precise claim can be wrong), completeness (individual claims don't indicate coverage), or importance (trivially true claims score high). The Q-0 calibration (Section 5.2) showed that BDI-specific criteria have different reliability profiles across models, so FIRE uses only the three universal criteria that are stable across all claim types.

4.3 Iterative Refinement Protocol

Claims below the confidence threshold enter a refinement loop. Each iteration sends a targeted prompt addressing one uncertain claim, asking the model to:

1. **Verify** the claim against source document text
2. **Cite** supporting passages as verbatim quotes
3. **Re-evaluate** evidence criteria in light of cited evidence
4. **Update** claim text if refinement reveals a more precise formulation

If the model returns `verified: false`, confidence is set to 0.1 and iteration stops — a strong signal of extraction hallucination.

Four termination guardrails prevent unbounded iteration: max 5 iterations per claim, max 20 iterations per document, 60-second wall-clock timeout, and 25 API calls per invocation. When any guardrail triggers, FIRE returns results accumulated so far with a `termination_reason` field.

4.4 The FIRE Sniff: Automatic Triggering

Not all documents benefit from iterative extraction. We implement a two-stage decision mechanism:

Stage 1 (pre-extraction, zero API cost): Three deterministic document-level signals trigger automatic FIRE: word count > 8,000 (context window degradation risk), chunked processing required (merge artifacts), or complex PDF layout (noisy conversion). If any signal fires, FIRE runs directly.

Stage 2 (post-extraction): For documents passing Stage 1, single-shot extraction runs first. Five output signals are evaluated: low-confidence claim rate > 30%, specificity collapse rate > 40%,

warrant deficit rate > 50%, unmapped concept rate > 40%, and claim clustering > 60%. If 2+ signals fire, FIRE re-runs on the document.

The cost model averages ~2.6 API calls per document (vs. 1.0 for single-shot, 8-25 for unconditional FIRE).

4.5 Evaluation (E1)

[RESULTS PENDING]

Methodology. We will construct a gold-standard claim set (PP-1) from 25 documents spanning all three POVs and varying in length (2,000-15,000 words), domain complexity, and document type (academic papers, policy reports, opinion pieces). Two annotators will independently identify factual claims and classify their evidence criteria. Inter-annotator agreement will be measured via Cohen's kappa.

Each document will be processed under three conditions: (1) single-shot extraction, (2) FIRE iterative extraction, and (3) FIRE with the two-stage sniff. Metrics: claim-level precision (are extracted claims real?), recall (are gold-standard claims found?), and F1. Additionally, we will compute mean absolute error on evidence criteria (specificity, has_warrant, internally_consistent) between AI-assigned and human-annotated values.

Expected results. We expect FIRE to improve precision (fewer hallucinated claims due to the verification loop) with modest recall improvement (targeted refinement may surface claims missed in the initial pass). The improvement should be largest on long, complex documents where single-shot extraction is most prone to specificity collapse and claim clustering.

5. QBAF: Formal Argument Strength in Multi-POV Discourse

5.1 DF-QuAD Gradual Semantics

We implement DF-QuAD (Rago et al., 2016) for computing argument strength from base scores and attack/support relationships. Given a set of arguments with base strengths b(a) in [0,1] and typed relationships (supports/attacks), DF-QuAD iteratively computes:

For each argument *a* with attackers *Att(a)* and supporters *Sup(a)*:

$$\sigma(a) = b(a) + b(a) \cdot \text{agg}(Sup(a)) - b(a) \cdot \text{agg}(Att(a))$$

where agg() aggregates the strengths of attacking or supporting arguments. The computation iterates until convergence (threshold: 0.001 absolute change in any argument's strength).

Attack types receive differential weights following Pollock's classification: *rebut* (weight 1.0, direct contradiction), *undercut* (weight 1.1, inference denial — slightly stronger because it challenges the reasoning, not just the conclusion), and *undermine* (weight 1.2, premise attack — strongest because it challenges the foundation). These weights are configurable.

The engine is pure computation — it does not depend on how base scores are sourced. This separation is critical: the calibration challenge (Section 5.2) affects base score quality but not the propagation mechanism.

5.2 BDI-Aware Base Score Calibration

The Q-0 calibration benchmark tests whether AI-assigned base scores correlate with human expert scores. Our calibration journey revealed a fundamental finding about the nature of different claim types.

Iteration 1: Holistic scoring. We prompted the AI to assign a single "how strong is this argument?" score (0-1). Result: Pearson r = -0.12 against human scores — essentially uncorrelated. The AI clustered all scores in the 0.7-0.9 range, failing to discriminate between weak and strong arguments.

Iteration 2: Decomposed rubric. Following evidence that LLMs perform better on checkable sub-criteria than holistic judgment (Wei et al., 2022), we decomposed "argument strength" into six boolean/categorical criteria: cites_source, source_quality, falsifiable, quantitative, source_consistent, and contested. Base strength was computed deterministically from criterion responses.

Discovery: BDI-blind bias. Analysis revealed that 4 of 6 criteria rewarded only empirical evidence (citations, source quality, falsifiability). A strong normative argument ("Nations should establish an AI safety body modeled on the IAEA, because catastrophic-potential technologies require multilateral oversight") scored at the 0.1 floor because it has no peer-reviewed source to cite, is not falsifiable, and contains no quantitative data. The rubric equated "argument strength" with "evidential grounding" — a category error that systematically underscored two of three BDI layers.

Iteration 3: BDI-aware rubric. We designed a rubric with three universal criteria (specificity, has_warrant, internally_consistent) and three BDI-specific criteria per layer:

- **Beliefs:** cites_source, source_quality, source_consistent
- **Desires:** values_grounded, tradeoff_acknowledged, precedent_cited
- **Intentions:** mechanism_specified, scope_bounded, failure_mode_addressed

Each BDI layer can achieve the same maximum score (1.0), evaluated on appropriate dimensions. The AI classifies the claim's BDI category first, then answers category-specific criteria.

Iteration 4: Calibration results. On 41 calibration claims (balanced across BDI layers) plus 8 hold-out claims:

Category	Scoring Method	Calibration r	Hold-out MAE
Beliefs	Human-assigned	N/A	N/A
Desires	AI rubric (v3)	0.65	0.13
Intentions	AI rubric (v3)	0.71	0.24

Desires and Intentions passed the calibration gate (r > 0.5 per-BDI, aggregate r > 0.7). Beliefs failed across all four prompt iterations.

5.3 Why Beliefs Resist Automated Scoring

The Beliefs calibration failure is not a prompt engineering problem — it reflects a fundamental asymmetry in what different BDI layers require for assessment:

Desires and Intentions claims are self-contained. Whether a normative argument is values-grounded, acknowledges tradeoffs, or cites precedent can be determined from the claim text alone. Whether a strategic claim specifies a mechanism or addresses failure modes is visible in the text. The AI's pattern recognition capabilities are well-suited to these assessments.

Beliefs claims require external verification. Whether an empirical claim accurately represents its cited source, whether the source is peer-reviewed, and whether the claim is consistent with the broader evidence base requires access to information outside the claim text — the actual source document, the journal database, the state of scientific consensus. The AI lacks reliable access to this external information and defaults to assessing surface features (assertive language, hedging words) rather than evidential quality.

This asymmetry generalizes beyond our system: any automated argumentation system that attempts to score argument strength must contend with the fact that empirical claims require fundamentally different assessment infrastructure than normative or strategic claims.

5.4 Evaluation (E2)

Calibration data. 41 claims scored by both human expert and AI rubric, balanced across BDI layers (~14 per category). Claims drawn from 15 source documents spanning AI policy, safety research, and economic analysis.

Metrics. Pearson correlation (r) for rank-order validation, Mean Absolute Error (MAE) for absolute accuracy. Per-BDI breakdowns to detect category-specific failure.

Results. The hybrid scoring approach (AI for Desires/Intentions, human for Beliefs) achieves the following aggregate calibration:

Metric	Desires	Intentions	Aggregate
Pearson r (calibration)	0.65	0.71	—
MAE (hold-out)	0.13	0.24	0.19

Hold-out validation (8 unseen claims) confirmed generalization. The per-BDI calibration gate ($r > 0.5$) passed for both AI-scored categories. Intentions MAE (0.24) is at the threshold — acceptable but flagged for monitoring in production use.

Failure analysis. AI scoring failures concentrated in three patterns: (1) implicit sources unrecognized (specific statistics like "77% of organizations" treated as uncited), (2) statistical warrants not counted as warrants (quantitative evidence IS the evidential ground, but the AI expected explicit "because" clauses), and (3) mechanism_specified applied inconsistently (critiques and diagnoses marked as mechanisms when they contain no alternative method). These patterns informed iterative prompt refinements (v1-v3).

6. Multi-Agent Debate with Ontological Grounding

6.1 Debate Protocol Design

Debates follow a structured protocol: (1) **clarification phase** — the moderator poses 2-3 scoping questions with multiple-choice options; the user selects answers that narrow the debate focus; (2) **topic refinement** — the moderator synthesizes answers into a precise debate question; (3) **opening statements** — each POV agent presents its initial position, grounded in its BDI-structured taxonomy context; (4) **cross-respond rounds** — the moderator selects which debater should respond to whom, on what specific point, based on argument network analysis; (5) **synthesis** — a separate AI pass extracts areas of agreement, disagreement, unresolved questions, and an AIF-aligned argument map with preference resolution.

Each debate agent receives its POV's taxonomy nodes organized by BDI category, with explicit framing headers. Primary nodes (most relevant to the debate topic, identified via embedding similarity) are marked with ⋆. Vulnerabilities known to be relevant to the topic are included with the instruction to acknowledge them when directly relevant — building credibility rather than over-conceding.

6.2 Dialectical Move Taxonomy and Diversity

We define six dialectical moves available to debate agents:

1. **COUNTEREXAMPLE** — provide a specific case challenging the opponent's general claim
2. **DISTINGUISH** — accept the opponent's evidence but show it doesn't apply to this context
3. **REFRAME** — shift the framing to reveal what the current frame hides
4. **REDUCE** — show the opponent's logic leads to an absurd or unacceptable conclusion
5. **ESCALATE** — connect the specific disagreement to a deeper principle
6. **CONCEDE** — acknowledge a valid point from the opponent

The rhetorical rigidity problem. In our initial implementation, 27 of 30 post-opening responses used the move pair [CONCEDE, DISTINGUISH], and every response began with the literal phrase "I concede." Analysis identified three compounding causes:

1. *Primacy bias in move ordering.* CONCEDE was listed first in the dialectical moves instruction block. LLMs disproportionately favor the first item in any enumerated list.
2. *JSON example anchoring.* The example JSON in the response prompt showed `"move_types": ["CONCEDE", "DISTINGUISH"]`, directly anchoring the model's output to this exact pair.
3. *No move diversity enforcement.* The model had no memory of its prior moves — it couldn't know it had already conceded three turns in a row.

Mitigation. We reordered moves (COUNTEREXAMPLE first, CONCEDE last), changed JSON examples to show varied moves, added explicit anti-repetition instructions ("Do NOT fall into a pattern of using the same moves every turn"), sentence variety guidance (alternative phrasings for concession), and move history injection (the model sees its last N moves and receives explicit instruction to vary).

6.3 Commitment Tracking and Concession Harvesting

Each debater maintains a commitment store tracking asserted, conceded, and challenged claims. The commitment store is injected into subsequent prompts with a consistency rule: "Do not silently contradict prior assertions."

Concession harvesting extends commitment tracking across debates. After synthesis, concessions are classified into three types using linguistic markers:

- **Full** (weight 1.0): Unconditional acceptance. Markers: "I accept that…", "You're right that…"
- **Conditional** (weight 0.5): Acceptance contingent on a condition. Markers: "I concede X, provided Y…"
- **Tactical** (weight 0.0): Arguendo concession. Markers: "Even if I accepted…", "For argument's sake…"

Classified concessions are accumulated per taxonomy node across debates. When weighted concession count on a node crosses a configurable threshold (default: 3.0 across 2+ distinct debates), the harvest dialog surfaces it as a candidate BDI update with three options: *qualify* (add caveat), *weaken* (reduce scope), or *retire* (archive as indefensible). All updates require human review — no automatic taxonomy changes.

This mechanism closes the loop between argumentation output and ontology evolution: repeated concessions signal that the taxonomy has drifted from defensible positions and needs revision. It also captures *convergence* — where POVs are coming together — complementing the situation nodes that capture where they diverge.

6.4 Preference Resolution and Synthesis

Debate synthesis evaluates which arguments prevail in each area of disagreement. Each preference judgment specifies the prevailing argument, the criterion by which it prevails (empirical_evidence, logical_validity, source_authority, specificity, or scope), and the rationale. This maps to AIF preference application nodes (PA-nodes).

The synthesis also produces an AIF-aligned argument map: claims with IDs, near-verbatim text, speaker attribution, and typed relationships (supported_by with scheme, attacked_by with attack_type and scheme). Schemes include COUNTEREXAMPLE, DISTINGUISH, REDUCE, REFRAME, and formal argumentation schemes (argument_from_evidence, argument_from_analogy, etc.).

6.5 Disagreement Typing

Each disagreement is classified into one of three types that map to BDI layers:

Type	BDI Layer	Resolvability	Example
EMPIRICAL	Belief divergence	Resolvable by evidence	"Does scaling eliminate bias?"
VALUES	Desire divergence	Negotiable via tradeoffs	"Is speed or safety more important?"
DEFINITIONAL	Conceptual divergence	Requires term clarification	"What counts as 'alignment'?"

This classification determines the appropriate resolution strategy: empirical disagreements call for evidence gathering, values disagreements call for tradeoff analysis, and definitional disagreements call for term disambiguation before substantive debate can proceed.

6.6 Evaluation (E3)

[RESULTS PENDING]

Methodology. We will conduct an A/B test comparing BDI-structured context injection against flat context injection. 20 debates will be generated on the same 10 topics: 10 with BDI-structured taxonomy context (nodes grouped by Beliefs/Desires/Intentions with framing headers, ⋆-tiered by relevance) and 10 with flat context (same nodes presented as an unstructured list without BDI grouping or relevance tiering).

Three human evaluators will rate each debate on four dimensions using a 5-point Likert scale: 1. **Argument quality:** Are claims well-structured (claim + evidence + warrant)? 2. **Taxonomy grounding:** Do agents reference taxonomy nodes appropriately and accurately? 3. **Disagreement identification:** Do agents correctly identify the type of their disagreements (empirical vs. values vs. definitional)? 4. **Perspective-taking:** Do agents engage with opposing viewpoints rather than talking past each other?

Inter-rater reliability will be measured via Krippendorff's alpha. Statistical significance will be assessed via paired t-test or Wilcoxon signed-rank test.

Expected results. We expect BDI-structured context to produce significantly higher scores on disagreement identification (agents taught the BDI framework should classify disagreements more accurately) and perspective-taking (agents seeing opposing interpretations via situation nodes should engage more directly). We expect modest improvement on argument quality (BDI framing encourages structured reasoning) and taxonomy grounding (⋆-tiering directs attention to relevant nodes).

7. Cross-POV Conflict Detection and Resolution

7.1 From Binary Matching to QBAF-Augmented Analysis

Early conflict detection used binary text matching: two claims are in conflict if they make contradictory assertions about the same topic. This approach produces high recall but low precision — superficially contradictory claims may actually address different aspects of the same topic, or the apparent contradiction may dissolve when the claims are understood in their BDI context.

QBAF-augmented conflict analysis adds formal argument strength to conflict instances. Instead of binary "these claims conflict," the system reports: these claims conflict, here are the attack/support relationships in the surrounding argument network, and here are the computed strengths after propagation. A conflict between a strong, well-supported claim (computed_strength 0.85) and a weak, isolated claim (computed_strength 0.3) is qualitatively different from a conflict between two equally supported positions — the former is likely resolvable by evidence, the latter represents a genuine crux.

7.2 Situation Nodes as Perspectival Anchors

DOLCE D&S situation nodes serve as shared reference points for cross-POV analysis. When two POVs interpret the same situation node differently, the `disagreement_type` classification (definitional, interpretive, or structural) identifies the nature of the divergence:

- **Definitional:** POVs define the key concept differently (e.g., "alignment" means different things to accelerationists and safetyists)
- **Interpretive:** POVs agree on the concept but disagree on its significance or implications
- **Structural:** POVs frame the concept within different causal or normative structures

This classification guides resolution strategy: definitional disagreements require term disambiguation, interpretive disagreements require evidence weighing, and structural disagreements require framework comparison.

7.3 Steelman Vulnerability and Fallacy Detection

Each taxonomy node carries optional `steelman_vulnerability` — the strongest counterargument to that position, expressed as a per-POV object describing how each opposing perspective would most effectively challenge it. This forces the system to represent positions at their strongest (steelmanning) before identifying weaknesses.

Fallacy detection classifies potential reasoning errors into four tiers: formal (logically invalid inference), informal_structural (argument scheme misapplication), informal_contextual (context-dependent error), and cognitive_bias (systematic reasoning distortion). Each detected fallacy includes a confidence level (likely/possible/borderline) and an explanation grounded in the specific node content, not generic critique.

8. Discussion

8.1 Vocabulary Over Formalism

A recurring design decision throughout this system is the adoption of ontological *vocabulary* — naming conventions, category tests, description patterns, and typed relationships — rather than formal ontological *reasoning* in OWL/RDF. We chose this approach because:

1. **The domain is discourse, not reality.** DOLCE was designed for how humans organize experience; BFO was designed for mind-independent reality. AI policy discourse is inherently perspectival — there is no "correct" ontological classification of whether "AI regulation should be proportional" is a Desire or an Intention. The category tests provide principled disambiguation without requiring formal subsumption reasoning.

2. **LLM prompts operate on vocabulary, not triples.** When we instruct a debate agent "YOUR EMPIRICAL GROUNDING (Beliefs): these are the factual claims you take as true," the ontological vocabulary shapes the agent's reasoning through natural language instruction. Encoding this as OWL triples would require a separate reasoning engine with no direct benefit to prompt quality.
3. **JSON structures are auditable and evolvable.** A taxonomy node with `category: "Desires"` and a genus-differentia description is immediately readable by human analysts. An OWL class hierarchy with rdfs:subClassOf relationships requires specialized tooling to inspect and modify.

This design choice has a cost: without formal reasoning, subsumption relationships, category membership, and constraint satisfaction must be enforced through prompt instructions and runtime validation (regex patterns, enum checks). For our use case — discourse analysis, not biomedical knowledge engineering — this tradeoff favors accessibility over formal completeness.

8.2 The Human-AI Scoring Boundary

The Q-0 calibration outcome reveals a principled boundary between what current LLMs can and cannot reliably assess. Normative and strategic claims (Desires, Intentions) are self-contained — their quality criteria (values grounding, tradeoff acknowledgment, mechanism specification) are visible in the claim text and assessable through pattern recognition, the LLM's core strength. Empirical claims (Beliefs) require external verification — checking source accuracy, evaluating evidence quality, and assessing consistency with the scientific record — which exceeds the model's reliable capabilities.

This finding generalizes: any automated argumentation system should expect different assessment reliability across different claim types, and the reliable/unreliable boundary will track the self-contained/externally-verifiable distinction, not the simple/complex distinction that might be assumed.

The hybrid approach (AI for Desires/Intentions, human for Beliefs) is not a failure of prompt engineering — it is a principled acknowledgment of the current human-AI task allocation boundary. As LLMs gain better grounding in external knowledge (through retrieval augmentation, tool use, or training data improvement), the boundary may shift. But the BDI-aware rubric design will remain useful: even with better models, separating the assessment criteria by claim type produces more auditable scores than holistic judgment.

8.3 Limitations

Taxonomy curation. While AI-assisted, the taxonomy requires significant human curation. The ~500 nodes represent a curated subset of AI policy discourse, not a comprehensive ontology. Coverage gaps exist, and the system's analytical quality depends on taxonomy quality.

Three-POV simplification. The accelerationist/safetyist/skeptic trichotomy is a deliberate simplification. Real AI policy discourse includes many more perspectives (industry, government, civil society, Global South, labor, etc.). The three-POV structure provides sufficient perspectival multiplicity for demonstrating the system's capabilities while remaining manageable.

English only. All documents, taxonomy nodes, and debate transcripts are in English. No cross-lingual evaluation has been conducted. The BDI framework and AIF vocabulary are language-independent, but the prompts, disambiguation tests, and genus-differentia patterns are English-specific.

FIRE evaluation gap. The gold-standard claim set (PP-1) required for rigorous FIRE evaluation has not yet been constructed. The FIRE methodology is presented based on system design and preliminary testing, not formal evaluation.

Embedding model. The all-MiniLM-L6-v2 model used for relevance scoring is a lightweight general-purpose encoder, not specialized for policy terminology. Domain-specific encoders (e.g., trained on policy/argumentation corpora) may improve relevance scoring accuracy.

Concession harvesting scale. The concession harvesting mechanism has been designed and implemented but not yet validated at scale. Meaningful validation requires concession data accumulated across 20+ debates — a volume that has not yet been reached.

8.4 Ethical Considerations

This system analyzes discourse about AI policy — a politically sensitive domain where computational tools can amplify certain perspectives while marginalizing others. Several ethical considerations apply:

Perspective selection bias. The three-POV structure (accelerationist, safetyist, skeptic) reflects perspectives prominent in Anglophone AI policy discourse. Perspectives from the Global South, indigenous communities, labor organizations, and non-technical stakeholders are not explicitly represented. The taxonomy's structure shapes which arguments are surfaced and which are invisible.

Analytical vs. advocacy framing. The system is designed as an analytical tool (mapping the discourse landscape) rather than an advocacy tool (recommending policy positions). However, the act of organizing perspectives into a structured taxonomy inevitably makes some framings more salient than others. The genus-differentia description format, by enforcing precision, may favor formalized academic perspectives over grassroots or experiential ones.

AI-generated debate content. Debate agents generate arguments based on taxonomy context, not personal belief or lived experience. Arguments attributed to the "safetyist" perspective are AI-generated approximations of safety-concerned positions, not authentic representations of any individual's views. This distinction must be clearly communicated to users.

Concession harvesting and evolving positions. Automated tracking of which positions are repeatedly conceded could be misused to claim that a perspective is "losing" the debate. The system explicitly frames concessions as signals about taxonomy accuracy, not about the merits of the underlying position — a distinction that must be maintained in any public presentation of results.

9. Conclusion

We have presented an integrated system for multi-perspective AI policy discourse analysis that addresses the fundamental limitation of flat, single-label stance detection. Our three-layer approach — ontological grounding (DOLCE D&S + BDI + AIF), formal argumentation (QBAF with BDI-aware calibration), and confidence-gated extraction (FIRE) — demonstrates that respecting the multi-dimensional structure of policy disagreements produces richer, more auditable analysis than compressing opinions into binary labels.

Our key findings include:

1. **BDI decomposition is principled and practical.** Separating empirical claims from normative commitments and strategic reasoning, using explicit disambiguation tests, improves both AI debate quality and argument strength calibration. The decomposition is not ad-hoc — it derives from established work in philosophy of mind and agent-based systems.
2. **Different claim types require different assessment infrastructure.** The Q-0 calibration outcome (AI succeeds on Desires/Intentions, fails on Beliefs) reveals a fundamental asymmetry traceable to the self-contained vs. externally-verifiable nature of different claim types. This finding generalizes beyond our system to any automated argumentation pipeline.
3. **Ontological vocabulary suffices for discourse analysis.** Adopting DOLCE/BDI/AIF naming conventions in JSON structures — without formal OWL/RDF reasoning — provides sufficient grounding to shape AI reasoning through prompt instructions and runtime validation. The "vocabulary over formalism" approach makes the system accessible and evolvable while preserving ontological rigor.
4. **Multi-agent debate requires active diversity management.** Without explicit move diversity enforcement, LLM debate agents converge on repetitive rhetorical patterns. Move ordering, JSON example selection, anti-repetition instructions, and move history injection are necessary prompt-level interventions.

Future work includes formal FIRE evaluation (E1), scaled concession harvesting validation, cross-lingual extension, and integration with retrieval-augmented generation to address the Beliefs scoring gap.

References

ALDayel, A. and Magdy, W. (2021). Stance detection with BERT embeddings for web discourse. *Proceedings of the 16th International AAAI Conference on Web and Social Media (ICWSM).*

Baroni, P., Rago, M., and Toni, F. (2019). From fine-grained properties to broad principles for gradual argumentation: A principled spectrum. *International Journal of Approximate Reasoning*, 105:252-286.

Bratman, M. (1987). *Intention, Plans, and Practical Reason.* Harvard University Press.

Cayrol, C. and Lagasquie-Schiex, M.-C. (2005). On the acceptability of arguments in bipolar argumentation frameworks. *Proceedings of the 8th European Conference on Symbolic and Quantitative Approaches to Reasoning with Uncertainty (ECSQARU)*, pages 378-389.

Chan, C. M., Chen, W., Su, Y., Yu, J., Xue, W., Zhang, S., Fu, J., and Liu, Z. (2024). ChatEval: Towards better LLM-based evaluators through multi-agent debate. *Proceedings of the 12th International Conference on Learning Representations (ICLR).*

Chesnevar, C. I., McGinnis, J., Modgil, S., Rahwan, I., Reed, C., Simari, G., South, M., Vreeswijk, G., and Willmott, S. (2006). Towards an argument interchange format. *The Knowledge Engineering Review*, 21(4):293-316.

Du, Y., Li, S., Torralba, A., Tenenbaum, J. B., and Mordatch, I. (2023). Improving factuality and reasoning in language models through multiagent debate. *Proceedings of the 40th International Conference on Machine Learning (ICML).*

Dung, P. M. (1995). On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming and n-person games. *Artificial Intelligence*, 77(2):321-357.

Ebrahimi, J., Dou, D., and Lowd, D. (2022). A survey of stance detection in online texts. *ACM Computing Surveys*, 54(3):1-37.

Guarino, N., Oberle, D., and Staab, S. (2009). What is an ontology? In *Handbook on Ontologies*, pages 1-17. Springer.

Janier, M. and Reed, C. (2014). OVA+: An argument analysis interface. *Proceedings of the 5th International Conference on Computational Models of Argument (COMMA).*

Jonnalagadda, S., Cohen, T., Wu, S., and Gonzalez, G. (2012). Enhancing clinical concept extraction with distributional semantics. *Journal of Biomedical Informatics*, 45(1):129-140.

Khan, A., Hughes, J., Valentine, D., Ruis, L., Sachan, M., and Perez, E. (2024). Debating with more persuasive LLMs leads to more truthful answers. *arXiv preprint arXiv:2402.06782.*

Lauscher, A., Ng, L., Napoles, C., and Tetreault, J. (2022). Rhetoric, logic, and dialectic: Advancing theory-based argument quality assessment. *Proceedings of the 29th International Conference on Computational Linguistics (COLING).*

Lawrence, J., Janier, M., and Reed, C. (2012). Auto-segmentation of dialogical argumentation. *Proceedings of the 4th Workshop on Computational Models of Natural Argument (CMNA).*

Li, Y., Sosea, T., Sawant, A., Nair, A. J., Inkpen, D., and Caragea, C. (2023). P-stance: A large dataset for stance detection in political domain. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL).*

Liang, T., He, Z., Jiao, W., Wang, X., Wang, Y., Wang, R., Yang, Y., Tu, Z., and Shi, S. (2023). Encouraging divergent thinking in large language models through multi-agent debate. *arXiv preprint arXiv:2305.19118.*

Ling, X. and Weld, D. S. (2012). Fine-grained entity recognition. *Proceedings of the 26th AAAI Conference on Artificial Intelligence.*

Luo, Y., Liu, Z., Shi, Y., and Zhang, Y. (2024). Exploring the sensitivity of LLMs to components of multi-dimensional stance. *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP).*

Masolo, C., Borgo, S., Gangemi, A., Guarino, N., and Oltramari, A. (2003). WonderWeb deliverable D18: Ontology library (final). *IST Project 2001-33052 WonderWeb*.

Mayer, T., Cabrio, E., and Villata, S. (2020). Transformer-based argument mining for healthcare applications. *Proceedings of the 24th European Conference on Artificial Intelligence (ECAI)*.

Mohammad, S. M., Kiritchenko, S., Sobhani, P., Zhu, X., and Cherry, C. (2016). SemEval-2016 task 6: Detecting stance in tweets. *Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval)*.

Rago, A., Toni, F., Aurisicchio, M., and Baroni, P. (2016). Discontinuity-free decision support with quantitative argumentation debates. *Proceedings of the 15th International Conference on Principles of Knowledge Representation and Reasoning (KR)*.

Rahwan, I., Zablith, F., and Reed, C. (2007). Laying the foundations for a world wide argument web. *Artificial Intelligence*, 171(10-15):897-921.

Rao, A. S. and Georgeff, M. P. (1991). Modeling rational agents within a BDI-architecture. *Proceedings of the 2nd International Conference on Principles of Knowledge Representation and Reasoning (KR)*.

Smith, B., Ashburner, M., Rosse, C., et al. (2015). The OBO Foundry: coordinated evolution of ontologies to support biomedical data integration. *Nature Biotechnology*, 25(11):1251-1255.

Stab, C. and Gurevych, I. (2017). Parsing argumentation structures in persuasive essays. *Computational Linguistics*, 43(3):619-659.

Törnberg, P. (2024). How to use LLMs for text analysis. *Proceedings of the National Academy of Sciences (PNAS)*, 121(24).

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., and Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems (NeurIPS)*, 35.